Entailment : (Question from exam - own solution)

Does the following entailment hold?

$$\{\exists x : F(x), \exists x, y : (R(x,y) \land G(y)), \forall x, y : ((F(x) \land R(x,y)) \to \neg G(y))\} \models \exists x, y : (F(x) \land R(x,y))$$

Give a first-order tableau proof or provide a countermodel.

$A \models B$ holds if $A \land \neg B$ is unsat (proved using tableau)

$\mathbf{t} : \exists x : F(x)$
$\mathbf{t} : \exists x, y : (R(x,y) \land G(y))$
$\mathbf{t} : \forall x, y : ((F(x) \land R(x,y)) \to \neg G(y))$
$\mathbf{f} : \exists x, y : (F(x) \land R(x,y))$

$\mathbf{t} : F(c_1)$
$\mathbf{t} : R(c_2, c_3) \land G(r_3)$
$\mathbf{t} : R(c_2, c_3)$
$\mathbf{t} : G(r_3)$

$\mathbf{t} : F(c_1) \land R(c_1, c_r) \to \neg G(r_3)$

$\mathbf{f} : F(c_1) \land R(c_1, c_3)$ | $\mathbf{t} : \neg G(c_3)$, $\mathbf{f} : G(c_3)$ ↯

$\mathbf{f} : F(c_1)$ ↯ | $\mathbf{f} : R(c_1, c_3)$

$\mathbf{f} : F(c_1) \land R(c_1, c_3)$
$\mathbf{f} : \ldots$ //remaining formulas derived from $\mathbf{f} : \exists x, y : F(x) \land R(x,y)$

$\mathbf{f} : F(c_1) \land R(c_1, c_3)$

$\mathbf{f} : F(c_1)$ ↯ | $\mathbf{f} : R(c_1, c_3)$

//no more inference rules can be applied to this branch and it didn't close
$\Longrightarrow$ negation of entailment IS satisfiable

Using the open branch, one can create the following counterexample to the entailment:

- $\mathcal{D} = \{c_1, c_2, c_3\}$ //Recall: the syntax of an intepreation is on the other page
- $I(F) = \{(c)\}$
- $I(G) = \{(c_3)\}$
- $I(R) = \{(c_2, c_3)\}$

Entailment : (Question from exam - own solution)

Consider the theory Γ and the two formulas φ_1 and φ_2

$\Gamma : \{\exists x : P(x), \forall x : (P(x) \to \forall y : R(x,y))\}$
$\varphi_1 : \forall x : (\exists y : R(x,y)) \quad \varphi_2 : \forall x : (\exists y : R(y,x))$

Which of the φ_i is not entailed by Γ? Exhibit a countermodel and argue that the interpretation you give is indeed a countermodel.

φ_1 is not entailed by Γ, as can bee seen given the interpretation I :

- $\mathcal{D} = \{a, b\}$
- $I(P) = \{a\}$
- $I(R) = \{(a,a), (a,b)\}$

Γ is satisfiable under I as there exists an element (a) for which $P(a)$ holds and that for every element x in the domain(a,b) $R(a,x)$ holds.
But it doesn't hold that for every element x in the domain (a,b) there exists an element y such that $R(x,y)$ holds, as can be seen for $x = a$.

//as for how one can see that $\Gamma \models \varphi_2$ using an informal proof:
Γ states that there exists some x for which $P(x)$ and subsequentially $R(x,y)$ for all y holds. This then means, that for every y there must exist at least one element (namely x) such that $R(x,y)$ holds.

FOL equivalences : ((Partial) Question from exercise sheet - multiple solutions)

Prove the following equivalence in FOL. You can use semantic arguments (e.g. arguing about all interpreations) or proof theory (e.g. tableau proofs).

$$\neg\forall x : \varphi \equiv \exists x : \neg\varphi$$

– **Own solution:**

$\neg\forall x : \varphi \equiv \exists x : \neg\varphi \iff G := (\neg\forall x : \varphi \to \exists x : \neg\varphi) \land (\exists x : \neg\varphi \to \neg\forall x : \varphi)$

The equivalence therefore holds, if $\neg G$ is unsat (provably using tableau)

$\mathbf{f} : (\neg\forall x : \varphi \to \exists x : \neg\varphi) \land (\exists x : \neg\varphi \to \neg\forall x : \varphi)$

$\mathbf{f} : \neg\forall x : \varphi \to \exists x : \neg\varphi$	$\mathbf{f} : \exists x : \neg\varphi \to \neg\forall x : \varphi$
$\mathbf{t} : \neg\forall x : \varphi$	$\mathbf{t} : \exists x : \neg\varphi$
$\mathbf{f} : \forall x : \varphi$	$\mathbf{t} : \neg\varphi(c_1/x)$
$\mathbf{f} : \varphi(c_1/x)$	$\mathbf{f} : \varphi(c_1/x)$
$\mathbf{f} : \exists x : \neg\varphi$	$\mathbf{f} : \neg\forall x : \varphi$
$\mathbf{f} : \neg\varphi(c_1/x)$	$\mathbf{t} : \forall x : \varphi$
$\mathbf{t} : \varphi(c_1/x)$	$\mathbf{t} : \varphi(c_1/x)$
↯	↯

Since the tableau for $\neg G$ is closed it has been proven unsat and therefore the equivalence has been proven to hold

– **Sample Solution:**

Let $\mathcal{M} = (\mathcal{D}, I)$ be a structure and $\ell : V \to \mathcal{D}$ a variable assignment for $\mathcal{M}$

$(\mathcal{M}, \ell) \models \neg\forall x : \varphi$
$\iff$ it is not that: $(\mathcal{M}, \ell) \models \forall x : \varphi$
$\iff$ it is not that: $(\mathcal{M}, \ell[d/x]) \models \varphi$, for all $d \in \mathcal{D}$
$\iff (\mathcal{M}, \ell[d/x]) \not\models \varphi$, exists $d \in \mathcal{D}$
$\iff (\mathcal{M}, \ell[d/x]) \models \neg\varphi$, exists $d \in \mathcal{D}$
$\iff (\mathcal{M}, \ell) \models \exists x : \neg\varphi$

Formalization (1) : (Question from exercise sheet - sample solution)

Formulize the following sentences:

1. All dogs howl at night.
2. Someone who has cat will not have any mice.
3. Light sleepers do not have anything which howls at night.
4. John has either a cat or a dog.

Using the constant *John* to represent John and the predicates

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- $Dog(x) :=$ x is a dog.
- $Howl(x) :=$ x howls at night.
- $Have(x,y) :=$ x has y.
- $Cat(x) :=$ x is a cat.
- $Mice(x) :=$ x is a mice.
- $LS(x) :=$ x is a light sleeper.

we can formulize the given sentences as:

1.$\forall x : (Dog(x) \to Howl(x))$
2.$\forall x, y : (Have(x,y) \land Cat(y) \to \neg\exists z : (Have(x,z) \land Mice(z)))$
3.$\forall x : (LS(x) \to \neg\exists y : (Have(x,y) \land Howl(y)))$
4.$\exists x : (Have(John, x) \land (Cat(x) \lor Dog(x)))$

Formalization (2) : (Question from exercise sheet - sample solution)

Formalize the following sentences in first-order logic:

1. 1. All actors and journalists invited to the party are late.
2. 2. There is at least one person on time.
3. There is at least one invited person who is neither a journalist nor an actor.

We use the predicates
- $Actor(x) :=$ x is an actor.
- $Journalist(x) :=$ x is a journalist.
- $Invited(x) :=$ x is invited
- $Late(x) :=$ x is late.

The above sentences can be formalised as follows

1.$\forall x : ((Actor(x) \lor Journalist(x)) \land Invited(x) \to Late(x))$
2.$\exists x : (\neg Late(x))$
3.$\exists x : (\neg(Actor(x) \lor Journalist(x)) \land Invited(x))$

Prenex Normal Form Transformation : (Question from exercise sheet - sample solution)

Convert the following FOL formulas into pnf:

$$\begin{aligned}
\psi_1 &: \quad \forall x(P(x) \to \exists y Q(x,y)) \land \exists x P(x) \\
\psi_2 &: \quad \exists x(P(x) \land \forall y(R(y) \to Q(x,y)) \land \neg(\exists y Q(y,x))) \\
\psi_3 &: \quad \neg\big(\forall x \exists y(\neg P(y) \lor Q(x, f(y)) \lor Q(y,z))\big)
\end{aligned}$$

- The case of ψ_1:
  1. Eliminate $\to$: $\forall x(\neg P(x) \lor \exists y Q(x,y)) \land \exists x P(x)$;
  2. Move all negations inwards to the atomic level (Nothing need to be done): $\forall x(\neg P(x) \lor \exists y Q(x,y)) \land \exists x P(x)$;
  3. Rename all variables apart: $\forall x_1(\neg P(x_1) \lor \exists x_2 Q(x_1,x_2)) \land \exists x_3 P(x_3)$;
  4. Move all quantifiers to the front:
     (a) $\forall x_1 \exists x_2(\neg P(x_1) \lor Q(x_1,x_2)) \land \exists x_3 P(x_3)$;
     (b) $\forall x_1(\exists x_2(\neg P(x_1) \lor Q(x_1,x_2)) \land \exists x_3 P(x_3))$;
     (c) $\forall x_1 \exists x_2((\neg P(x_1) \lor Q(x_1,x_2)) \land \exists x_3 P(x_3))$;
     (d) $\forall x_1 \exists x_2 \exists x_3((\neg P(x_1) \lor Q(x_1,x_2)) \land P(x_3))$.
- The case of ψ_2:
  1. Eliminate $\to$: $\exists x(P(x) \land \forall y(\neg R(y) \lor Q(x,y)) \land \neg(\exists y Q(y,x)))$
  2. Move all negations inwards to the atomic level: $\exists x(P(x) \land \forall y(\neg R(y) \lor Q(x,y)) \land \forall y \neg Q(y,x))$;
  3. Rename all variables apart: $\exists x_1(P(x_1) \land \forall x_2(\neg R(x_2) \lor Q(x_1,x_2)) \land \forall x_3 \neg Q(x_3,x_1))$;
  4. Move all quantifiers to the front:
     (a) $\exists x_1 \forall x_2(P(x_1) \land (\neg R(x_2) \lor Q(x_1,x_2)) \land \forall x_3 \neg Q(x_3,x_1))$;
     (b) $\exists x_1 \forall x_2 \forall x_3(P(x_1) \land (\neg R(x_2) \lor Q(x_1,x_2)) \land \neg Q(x_3,x_1))$.
- The case of ψ_3:
  1. Eliminate $\to$ (Nothing need to be done): $\neg\big(\forall x \exists y(\neg P(y) \lor Q(x,f(y)) \lor Q(y,z))\big)$;
  2. Move all negations inwards to the atomic level:
     (a) $\exists x \forall y \neg(\neg P(y) \lor Q(x,f(y)) \lor Q(y,z))$;
     (b) $\exists x \forall y(P(y) \land \neg Q(x,f(y)) \land \neg Q(y,z))$;
  3. Rename all variables apart: $\exists x_1 \forall x_2(P(x_2) \land \neg Q(x_1,f(x_2)) \land \neg Q(x_2,x_3))$;
  4. Move all quantifiers to the front (Nothing need to be done): $\exists x_1 \forall x_2(P(x_2) \land \neg Q(x_1,f(x_2)) \land \neg Q(x_2,x_3))$.

Theory consistency : (Question from exercise sheet - sample solution)

Consider the following formulas.

$$\varphi_1 = \forall x \exists y R(x,y) \qquad \varphi_2 = \forall x \forall y(R(x,y) \to R(y,x))$$
$$\psi_1 = \exists x \forall y R(x,y) \qquad \psi_2 = \forall x \forall y \forall z((R(x,y) \land R(y,z)) \to R(x,z)).$$

Answer the questions.

(a) Is the theory $\{\varphi_1, \varphi_2, \psi_1\}$ consistent? (or equivalently, is $\varphi_1 \land \varphi_2 \land \psi_1$ satisfiable?)

(b) What about the theory $\{\neg\varphi_1, \varphi_2, \psi_1\}$, is it consistent?

(c) Finally, consider the theory $\{\varphi_1, \varphi_2, \neg\psi_1, \neg\psi_2\}$. Is it consistent?

(a) Let $\mathcal{M} = (D, I)$ be a structure and $\ell : V \to D$ a variable assignment for $\mathcal{M}$ that

- $D = \{0,1\}$ and $R^{\mathcal{M}} = I(R) = \{(0,0),(0,1),(1,0)\}$.

So $\{\varphi_1, \varphi_2, \psi_1\}$ is satisfiable in $\mathcal{M}$ wrt ℓ:

- $R^{\mathcal{M}}$ is serial and symmetric, so φ_1 and φ_2 are satisfiable in $\mathcal{M}$ wrt ℓ;
- ψ_1 is satisfiable — there is a $0 \in D$ such that $(0,1)$ and $(0,0)$ are both in $R^{\mathcal{M}}$.

(b) Suppose $\{\neg\varphi_1, \varphi_2, \psi_1\}$ is satisfiable in $\mathcal{M} = (D, I)$ wrt ℓ that $R^{\mathcal{M}} = I(R)$.

- From $\neg\varphi_1$ is satisfiable, there is a $d \in D$ for all $c \in D$ that $(d,c) \notin R^{\mathcal{M}}$ (@$_1$).
- From ψ_1 is satisfiable, there is a $d' \in D$ for all $c \in D$ that $(d',c) \in R^{\mathcal{M}}$ (@$_2$).
- From φ_2 is satisfiable in $\mathcal{M}$ wrt ℓ, $R^{\mathcal{M}}$ is symmetric. In addition with (@$_2$), which gives $(d',d) \in R^{\mathcal{M}}$, so $(d,d') \in R^{\mathcal{M}}$. However, this result is in conflict to (@$_1$).

So $\{\neg\varphi_1, \varphi_2, \psi_1\}$ is **not** satisfiable.

(c) Let $\mathcal{M} = (D, I)$ be a structure and $\ell : V \to D$ a variable assignment for $\mathcal{M}$ that

- $D = \{0,1\}$ and $R^{\mathcal{M}} = I(R) = \{(0,1),(1,0)\}$.

So $\{\varphi_1, \varphi_2, \neg\psi_1, \neg\psi_2\}$ is satisfiable in $\mathcal{M}$ wrt ℓ:

- As $R^{\mathcal{M}}$ is serial and symmetric, φ_1 and φ_2 are satisfiable in $\mathcal{M}$ wrt ℓ;
- From $(0,0),(1,1) \notin R^{\mathcal{M}}$, we know $\neg\psi_1$ is satisfiable in $\mathcal{M}$ wrt ℓ;
- From $R^{\mathcal{M}}$ is not transitive — $(0,1),(1,0) \in R^{\mathcal{M}}$ but $(0,0) \notin R^{\mathcal{M}}$, it follows that $\neg\psi_2$ is satisfiable in $\mathcal{M}$ wrt ℓ.

Finding a model : (Question from exercise sheet - sample solution)

Consider the following formulas.

$\varphi_1 = \forall x : R(x, s(x))$
$\varphi_2 = \forall x, y : (R(x,y) \to \neg R(y,x))$
$\varphi_3 = \forall x, y, z : (R(x,y) \land R(y,z) \to R(x,z))$

a) Can you find a model of $\varphi_1 \land \varphi_2 \land \varphi_3$?
b) Is there a model of $\varphi_1 \land \varphi_2 \land \varphi_3$ with a finite domain? Explain your answer!

(a) Let $\mathcal{M} = (D, I)$ be a structure and $\ell : V \to D$ a variable assignment for $\mathcal{M}$ that

- $D = \mathbb{N}$ and $I(R) =<$, as well as $\mathsf{s} = I(s) : \mathbb{N} \to \mathbb{N}$ such that $\mathsf{s}(n) = n+1$ where $n \in \mathbb{N}$.

So $\varphi_1 \land \varphi_2 \land \varphi_3$ is satisfiable in $\mathcal{M}$:

- φ_1 is satisfiable in $\mathcal{M}$ wrt ℓ — $(n, n+1) \in <$ for all $n \in \mathbb{N}$.
- As $<$ is asymmetric and transitive, therefore the model $\mathcal{M}$ satisfies φ_2 and φ_3 as well.

(b) Suppose there is a finite model $\mathcal{M} = (D, I)$ and $\ell : V \to D$ a variable assignment for $\mathcal{M}$ such that $D = \{d_1, \ldots, d_n\}$, $R^{\mathcal{M}} = \mathsf{R} \subseteq D \times D$, and $s^{\mathcal{M}} = \mathsf{s} : D \to D$. Assume that $\{\varphi_1, \varphi_2, \varphi_3\}$ is satisfiable in $\mathcal{M}$ wrt ℓ. So,

- From $\mathcal{M}, \ell \models \varphi_1$ it follows that for all $d \in D$ that $d\mathsf{R}\mathsf{s}(d)$ (@$_1$);
- From $\mathcal{M}, \ell \models \varphi_2$ it follows that for all $d, c \in D$ that if $d\mathsf{R}c$ then **not** -$c\mathsf{R}d$ (@$_2$).
- From $\mathcal{M}, \ell \models \varphi_3$ it follows that for all $d, c, e \in D$ that if $d\mathsf{R}c$ and $c\mathsf{R}e$ then $d\mathsf{R}e$ (@$_3$).
- We define a new notation $\mathsf{s}^{n+1}(d) = \overbrace{\mathsf{s}(\ldots\mathsf{s}}^{n+1}(d)) = \mathsf{s}(\mathsf{s}^n(d))$ — so, $\mathsf{s}^n(d)$ means that we apply the function s on $d \in D$ in n times.
- By (@$_1$), it follow that $\mathsf{s}^n(d)\mathsf{R}\mathsf{s}^{n+1}(d)$ where $n \in \mathbb{N}$ (@$_4$) — meaning that every $\mathsf{s}^n(d)$ is accessed by its direct successor $\mathsf{s}^{n+1}(d)$.
- By (@$_3$) and (@$_4$), we have $d\mathsf{R}\mathsf{s}^k(d)$ (@$_5$) — meaning that every d is accessed by its k-successor $\mathsf{s}^k(d)$, where $k \in \mathbb{N}$.
- By (@$_2$), given any dRc, it is not possible to have $d = c$. Otherwise, it would lead to dRd which is in conflict to (@$_2$). This means that, every point cannot be its successor nor its ancestor (@$_6$).
- By (@$_5$) and (@$_6$), we have $d\mathsf{R}\mathsf{s}^1(d), d\mathsf{R}\mathsf{s}^2(d), \ldots, d\mathsf{R}\mathsf{s}^n(d)$; however, $d, \mathsf{s}^1(d), \mathsf{s}^2(d), \ldots, \mathsf{s}^n(d)$ are $n+1$ different points, which is contrary to the assumption that D only have n elements!

- From (@$_4$), we can enumerate $d\mathsf{R}\mathsf{s}^1(d), \mathsf{s}^1(d)\mathsf{R}\mathsf{s}^2(d), \ldots, \mathsf{s}^{k-1}(d)\mathsf{R}\mathsf{s}^k(d)$, where $k < n$.
- Otherwise, $k \geq n$. Suppose $k = n$ and so $\mathsf{s}^{n-1}(d)\mathsf{R}\mathsf{s}^n(d)$.
- Given that D only have n elements, there must be a $i < n$ such that $\mathsf{s}^n(d) = \mathsf{s}^i(d)$. So $\mathsf{s}^{n-1}(d)\mathsf{R}\mathsf{s}^i(d)$ (@$_7$).
- By (@$_3$) and the enumeration, we have $\mathsf{s}^i(d)\mathsf{R}\mathsf{s}^{n-1}(d)$ (@$_8$).
- However, (@$_7$) and (@$_8$) together are in conflict to (@$_2$).